

Proteinase and exopeptidase hydrolysis of whey protein: Comparison of the TNBS, OPA and pH stat methods for quantification of degree of hydrolysis

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Abstract

Whey protein hydrolysates were generated with Alcalase 2.4L and Debitrase HYW20 which are proteinase and exopeptidase enriched enzyme preparations, respectively. Degree of hydrolysis (DH) values were quantified with the TNBS, OPA and pH stat methods. Poor correlation was observed between the three methods for DH values in Debitrase HYW20 hydrolysates. For Alcalase 2.4L hydrolysates, the OPA method gave DH values that were approximately 15% lower than the pH stat, whereas TNBS DH values were similar to the pH stat method. As whey proteins are relatively rich in cysteine, a weak and unstable reaction between OPA and cysteine was thought to contribute to the under-estimation of DH in whey protein hydrolysates. Since TNBS reacts strongly with cysteine and TNBS DH values were unaffected by the type of enzyme preparation used to generate the hydrolysate, the TNBS method was deemed most suitable for the quantification of DH in whey protein hydrolysates.

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1. Introduction

Enzymatic hydrolysis has been used for centuries for modification of the functional and nutritional properties of food proteins in the production of traditional foods such as cheeses and fermented plant foods. Enzymatic proteolysis has been shown to increase solubility, modify foaming, emulsifying and gelation properties and to liberate biologically active peptides from certain proteins.

However, many authors have shown that the extent to which the functional properties of a protein may be altered by hydrolysis is very much dependant on the degree to which the protein has been hydrolysed. Obtaining a value for the actual number of peptide bonds cleaved during the reaction, or degree of hydrolysis (DH) is a useful way of monitoring the extent of protein degradation. DH is defined as the percentage of the total number of peptide bonds in a

protein which have been cleaved during hydrolysis (Adler-Nissen, 1986).

Limited hydrolysis has been shown to improve both foaming and emulsifying properties of proteins (Keuhler & Stine, 1974; Adler-Nissen & Olsen, 1979; Lieske & Konrad, 1996; Ipsen et al., 2001). Lieske and Konrad (1996) concluded that the optimal DH for the improvement of functionality was 3.0%. More extensive hydrolysis has been shown to increase solubility (Acchouri, Zhang, & Shiyang, 1998; Slattery & FitzGerald, 1998; Flanagan & FitzGerald, 2002) and modify the gelation properties of food proteins (Ju & Kilara, 1998; Otte, Schumacher, Ipsen, Ju, & Qvist, 1999; Doucet, Gauthier, & Foegeding, 2001).

Numerous methods exist for the estimation of DH. It can be quantified by determining the amount of nitrogen released during hydrolysis, which becomes soluble in the presence of a precipitating agent (e.g. trichloroacetic acid). Methods used include the Kjeldahl method (AOAC, 1995) or spectrophotometric determination in the visible region after colorimetric reaction e.g. biuret reaction (Hung, Vas, Cheke, & Bolcsi, 1984). DH can also be quantified by determination of the free amino

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groups released during hydrolysis by formol titration (Juffs, 1975), or by using compounds which react specifically with amino groups such as trinitrobenzenesulphonic acid (TNBS) (Okuyama & Satake, 1960; Habeeb, 1966; Fields, 1971; Adler-Nissen, 1979a; Polychroniadou, 1988; Humbert, Guingamp, Kouomegne, & Linden, 1990; Caer & Colas, 1993), *o*-phthalaldehyde (OPA) (Church, Porter, Catignani, & Swaisgood, 1985; Frister, Meisel, & Schlimme, 1986; Garcia Alvarez-Coque, Medina Hernandez, Villaneuva Camanas, & Mongay Fernandez, 1989a,b; Medina Hernandez, Villaneuva Camanas, Monfort Cuenca and Garcia Alvarez-Coque, 1990a,b, 1991; Nielsen, Petersen, & Dambmann, 2001), ninhydrin (Pearce, Karahalios & Friedman, 1988) or fluorescamine (Pearce, 1979). DH can also be quantified using osmometry (Adler-Nissen, 1986), where the depression in freezing point can be used to calculate changes in osmolality during hydrolysis, which is then used to calculate DH. Alternatively, in the pH stat method the protons released during hydrolysis are titrated and then related to DH (Adler-Nissen, 1986; Margot, Flaschel, & Renken, 1994; Camacho, Gonzalez-Tello, Paez-Duenas, Guadix, & Guadix, 2001). However, the three approaches most commonly employed for the quantification of DH during food protein hydrolysis are the pH stat, OPA and TNBS methods.

The principle of the pH stat technique is that when hydrolysis is carried out at neutral or alkaline conditions, dissociation of protons from the free amino groups released is favoured (Adler-Nissen, 1986). The liberation of protons into the surrounding medium leads to a reduction in the pH of the reaction mixture. The number of peptide bonds cleaved can be estimated from the amount of base required to maintain a constant pH during the reaction (Adler-Nissen, 1986).

The OPA method is based on the specific reaction between OPA and primary amino groups, in the presence of a thiol to form 1-alkylthio-2-alkyl-substituted isoindoles (Medina Hernandez et al., 1990a,b). The isoindoles formed can be quantified spectrophotometrically at 340 nm or fluorometrically at 455 nm.

TNBS also reacts specifically with primary amino groups to form a chromophore with a maximum absorbance at 340 nm (Adler-Nissen, 1979a).

The objective of this study was to compare the OPA, TNBS and pH stat methods for quantification of the DH in whey protein hydrolysates generated with an enzyme preparation containing mainly proteinase activities (Alcalase 2.4L) and an exopeptidase rich enzyme preparation (Debitrase HYW20).

2. Materials and methods

Whey protein concentrate (WPC 75) was purchased from a commercial supplier. Di-sodium hydrogen

orthophosphate di-hydrate, sodium di-hydrogen orthophosphate di-hydrate, L-proline, L-serine, L-glycine, L-alanine, L-leucine, L-cysteine, L-methionine, L-phenylalanine, L-trptophan, L-histidine, L-aspartic acid and L-glutamic acid were obtained from BDH (Poole, Dorset, UK). Boric acid, NaOH, OPA, *N*-acetyl-L-cysteine (NAC), sodium dodecyl sulphate (SDS), picrylsulphonic acid (TNBS), L-arginine, L-valine, L-isoleucine, L-threonine, L-asparagine, L-glutamine, L-tyrosine and L-lysine were supplied by Sigma Chemical Co. (Poole, Dorset, UK).

Debitrase HYW20[®] was kindly supplied by Rhodia Ltd. (Cheshire, UK), Alcalase 2.4L[®] was from Novo Nordisk A/S (Bagsvaerd, Denmark).

All other reagents were of analytical grade, unless otherwise stated.

2.1. Enzymatic hydrolysis of WPC

Hydrolysis experiments were carried out in a 500 mL sealed reaction vessel (Metrohm ion analysis, Herisau, Switzerland). A 20 g 100 mL⁻¹ aqueous solution of whey protein concentrate (73.96% (w/w) protein) was allowed to hydrate for 1 h at room temperature with gentle mixing. The protein solution was then equilibrated at 50°C and the pH adjusted to 7.0 with 2.0 N NaOH before addition of the enzyme. Debitrase HYW20 was added at an enzyme: substrate (E:S) ratio of 1% (weight of enzyme powder:weight of protein), and Alcalase 2.4L was added at a rate of 0.25 mL 100 mL⁻¹ protein solution. The solution was agitated with an over-head stirrer (Heidolph Instruments, Schwabach, Germany) throughout hydrolysis. Hydrolysate samples (50 µL) were taken at various time intervals, added to 950 µL of 5% (w/v) SDS, and then heated at 85°C, for 5 min to inactivate the enzyme activities. Samples were then stored at -20°C until required for analysis.

2.2. pH stat method for quantification of DH

Quantification of DH by the pH stat method was carried out using data obtained with a 718 Stat Titrimo (Metrohm ion analysis, Herisau, Switzerland). The base used for maintaining constant pH was 2.0 N NaOH. The percentage DH was calculated using the following formula (Adler-Nissen, 1986):

$$\text{DH}\% = 100BN_b(1/\alpha)(1/\text{MP})(1/h_{\text{tot}}), \quad (1)$$

where B is the base consumption in mL, N_b the normality of the base, α the average degree of dissociation of the α -NH₂ groups, MP the mass of protein being hydrolysed (g), and h_{tot} the total number of peptide bonds in the protein substrate (meq g⁻¹ protein).

The degree of dissociation (α) for the α -NH₂ groups was calculated as follows:

$$\alpha = \frac{10^{(\text{pH}-\text{pK})}}{1 + 10^{(\text{pH}-\text{pK})}}, \quad (2)$$

where pK is the average dissociation value for the α -amino groups liberated during hydrolysis and is dependant on temperature, peptide chain length and the nature of the terminal amino acid. At 50°C (the hydrolysis temperature used in the present study), the average pK value for the α -amino groups of peptides and proteins is 7.1 (Adler-Nissen, 1986). At pH 7.0 (the pH value for all hydrolysis reactions) α can be calculated as 0.44, therefore the $1/\alpha$ value used was 2.27. The parameter h_{tot} is given as meqv peptide bonds per gram of protein. This was calculated from amino acid analysis by summing the mmoles of each individual amino acid per gram of protein. The h_{tot} for whey protein concentrate is 8.8 meqv per g protein (Adler-Nissen, 1986).

2.3. Quantification of DH using OPA

The OPA method used to measure DH was a modification of that of Church, Swaisgood, Porter, and Catignani (1983). This involved using NAC as the thiol reagent, as described by Garcia Alvarez-Coque et al. (1989a,b).

The OPA/NAC reagent (100 mL) was prepared by combining 10 mL of 50 mM OPA (in methanol) and 10 mL of 50 mM NAC, 5 mL of 20% (w/v) SDS, and 75 mL of borate buffer (0.1 M, pH 9.5). The reagent was covered with aluminium foil to protect from light and allowed to stir for at least 1 h before use.

The OPA assay was carried out by the addition of 20 μ L of sample (or standard) to 2.4 mL of OPA/NAC reagent. The absorbance of this solution was measured at 340 nm with an Ultraspec 2000 (Pharmacia Biotech, Cambridge, England). The absorbance values for the interaction of amino groups with OPA were taken after 2 min standing for unhydrolysed WPC and after 10 min standing for hydrolysed WPC, i.e., after absorbance values at 340 nm had reached a plateau. In the case of the interaction of OPA with cysteine, the absorbance taken was the maximum absorbance value reached during the time course of incubation with OPA, i.e., after approx. 5 min. A standard curve was prepared using L-isoleucine (0–2 mg mL⁻¹).

DH values were calculated using the following formula:

$$\text{DH}\% = \frac{100n}{N}, \quad (3)$$

where n is the average number of peptide bonds hydrolysed, N the total number of peptide bonds per protein molecule

It can be shown that:

$$n = \frac{\Delta\text{Abs}Md}{\epsilon c}, \quad (4)$$

where ΔAbs is the Abs of test sample at 340 nm – Abs unhydrolysed sample at 340 nm, M the molecular mass of the test protein (Da), d the dilution factor ϵ the molar extinction coefficient at 340 nm (6000 mol⁻¹ cm⁻¹) (Church et al., 1985), and c the protein concentration (g L⁻¹).

As WPC contains a heterogeneous mixture of many different proteins an estimate was made of the average number of peptide bonds per mole of WPC and the average molecular mass of the proteins in WPC. Values were calculated taking the approximate protein content of WPC to be 62.5% β -lactoglobulin, 25% α -lactalbumin and 12.5% bovine serum albumin. Using these values the average molecular mass of the proteins in WPC was calculated to be 23644 Da (M), with an average of 204 peptide bonds per protein molecule (N).

By combining Eqs. (3) and (4) and inserting the calculated values for M and N , the following equation can be obtained:

$$\text{DH}\% = 100 \left[\left(\frac{\Delta\text{Abs}23644d}{6000c} \right) / 204 \right],$$

therefore

$$\text{DH}\% = \frac{\Delta\text{Abs}1.934d}{c}. \quad (5)$$

This formula (Eq. (5)) was then used for all subsequent calculations of DH by the OPA method.

2.4. Quantification of DH using TNBS

The method used was that described by Adler-Nissen (1979a). The TNBS reagent consisted of 0.1% (w/v) TNBS in water. All samples and standard solutions were prepared in 1% (w/v) SDS.

Duplicate aliquots (0.25 mL) of test or standard solutions were added to test tubes containing 2.0 mL of sodium phosphate buffer (0.2125 M, pH 8.2). TNBS reagent (2.0 mL) was then added to each tube, followed by mixing and incubation at 50°C for 60 min in a covered water bath (to exclude light). After incubation, the reaction was stopped by the addition of 0.1 N HCl (4.0 mL) to each tube. Samples were then allowed to cool at room temperature for 30 min, before absorbance values were measured at 340 nm using an Ultraspec 2000 (Pharmacia Biotech, Cambridge, England).

L-Leucine (0–2.0 mM) was used to generate a standard curve. DH values were calculated using the following formula (Adler-Nissen, 1979b):

$$\text{DH}\% = 100 \left(\frac{\text{AN}_2 - \text{AN}_1}{\text{Npb}} \right), \quad (6)$$

where AN_1 is the amino nitrogen content of the protein substrate before hydrolysis (mg g^{-1} protein), AN_2 the amino nitrogen content of the protein substrate after hydrolysis (mg g^{-1} protein), and N_{pb} the nitrogen content of the peptide bonds in the protein substrate (mg g^{-1} protein). A value of 123.3 was used for whey protein (Adler-Nissen, 1979b).

The values of AN_1 and AN_2 were obtained by reference to a standard curve of Abs at 340 nm versus mg L^{-1} amino nitrogen (generated with L-leucine). These values were then divided by the protein content of the test samples to give mg amino nitrogen per gram of protein.

3. Results and discussion

The three methods (OPA, TNBS and pH stat) were first used to study the hydrolysis of WPC75 over a 6 h incubation period with Debitrase HYW20. The DH values obtained by the three methods are shown in Fig. 1.

The DH values obtained by the three methods were considerably different, with values for TNBS > OPA > pH stat. After 6 h hydrolysis, DH values of 19.3%, 16.8% and 12.3% were obtained from the TNBS, OPA and pH stat methods, respectively. The underestimation of DH by the pH stat method was as expected, since Debitrase HYW20 is enriched with exopeptidase activities (Rhodia Inc. Technical Bulletin: TB DEB 99-11.0). This underestimation is due to the fact that the pK value of the α -amino group is considerably higher for tripeptides, dipeptides and free amino acids, than for polypeptides (Adler-Nissen, 1986). This leads to an under-estimation of the value for $1/\alpha$ used in the calculation of DH by the pH stat method (see Eq. (1)), and a consequent under-estimation of DH.

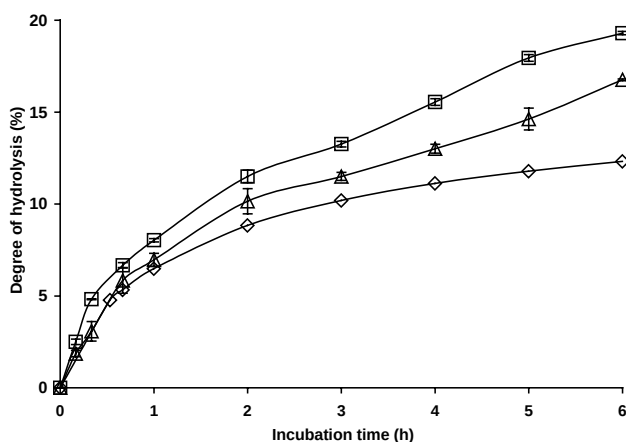


Fig. 1. DH (%) values obtained by the pH stat ($\diamond$), TNBS ($\square$) and OPA ($\triangle$) methods for the hydrolysis of a 20% (w/v) WPC 75 with Debitrase HYW20 over 6 h, at 50°C, pH 7.0. Values are means of duplicate determinations. Error bars show standard deviation.

For example, if the average pK value for the α -amino groups of polypeptides and proteins is 7.1 (at 50°C) and the average pK value for the α -amino groups of di- and tripeptides is approximately 0.5 pH unit higher (Adler-Nissen, 1986), then α values of 0.44 and 0.20 can be calculated for a mixture of polypeptides and a mixture of di- and tripeptides, respectively, at pH 7.0 (see Eq. (2)). Therefore, the $1/\alpha$ value calculated for a mixture of polypeptides and proteins is 2.27 and the $1/\alpha$ value calculated for a mixture of di- and tripeptides is 5.00 (50°C, pH 7.0). DH values calculated by the pH stat method for hydrolysates generated with exopeptidase-rich preparations would therefore be lower than values obtained by the TNBS and OPA methods, as the hydrolysate would contain relatively high amounts of free amino acids. However, the OPA and TNBS methods both quantify the amount of free amino groups released during hydrolysis and DH values obtained by these two methods would be expected to be similar. However, the results in Fig. 1 show that one, or both of these methods gave inaccurate results. To determine which method gave the most accurate DH values, it was decided to hydrolyse WPC75 with Alcalase 2.4L, a proteinase preparation. When using a proteinase such as Alcalase 2.4L to generate hydrolysates with moderate DH values (i.e. below 20%), the relative content of free amino acids and dipeptides is likely to be low (Adler-Nissen, 1986). This means that the change in the $1/\alpha$ value during hydrolysis should be negligible, and DH values obtained by the pH stat would be accurate. It should, therefore, be possible to compare DH values obtained by the pH stat method to those obtained by OPA and TNBS, to determine which of these two methods gave the most accurate value for DH.

Fig. 2 shows DH values obtained by the pH stat, OPA and TNBS methods for the hydrolysis of WPC75 with Alcalase 2.4L. The DH values obtained by the TNBS

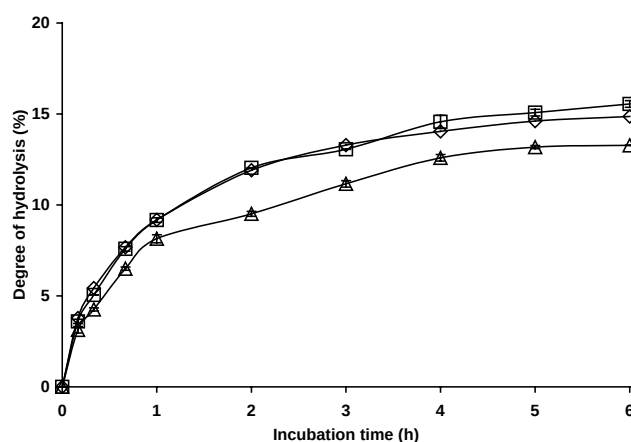


Fig. 2. DH (%) values obtained by the pH stat ($\diamond$), TNBS ($\square$) and OPA ($\triangle$) methods for the hydrolysis of a 20% w/v WPC75 solution with Alcalase 2.4L over 6 h, at 50°C, pH 7.0. Values are means of duplicate determinations. Error bars show standard deviations.

and pH stat methods agreed very well, but the values obtained by the OPA method were considerably lower. After 6 h hydrolysis, DH values of 15.7, 13.3, and 14.9 were obtained by the TNBS, OPA and pH stat methods, respectively. Over the 6 h hydrolysis period, DH values obtained by the OPA method were on average 14.68% lower than pH stat values; however, the TNBS values were on average only 0.03% higher than pH stat values. This underestimation of DH by the OPA method concurs with the difference between values obtained by the two methods for the hydrolysis of WPC75 with Debitrase HYW20, where the OPA values were on average 17.81% lower than TNBS values over the 6 h hydrolysis period (Fig. 1).

Chen, Scott, and Trepman (1979), on studying the fluorescence properties of OPA derivatives of amino acids, reported that OPA and cysteine react weakly, to give an unstable product. The weak reaction between OPA and cysteine has since been reported by many authors, in both fluorometric and spectrophotometric studies (Svedas, Galaev, Borisov, & Berezin, 1980; Garcia Alvarez-Coque et al., 1989a, b; Medina Hernandez et al., 1990a, b; Turgeon, Bard, & Gauthier, 1991; Silvestre, 1997). It would appear that the sulphhydryl group of cysteine is responsible for the weak reaction, as it competes intramolecularly with NAC for position 1 in the isoindole (Garcia Alvarez-Coque et al., 1989a, b). Cysteine reacts normally with OPA if its sulphhydryl group is blocked (Chen et al., 1979). This weak and unstable reaction between OPA and cysteine is a possible reason for the underestimation of DH in whey protein hydrolysates by the OPA method, as whey proteins are relatively rich in cysteine e.g. α -lactalbumin has a cysteine content of 6.5% (Swaisgood, 1982). The reaction between TNBS and cysteine is comparable to the reaction of TNBS with the other amino acids (except lysine and proline), which would explain why the TNBS method is unaffected by the relatively high cysteine content of whey proteins. The extinction coefficient (ϵ) values for the reactions between both OPA and TNBS and 20 amino acids were determined and are summarised in Table 1. Table 1 shows that the ϵ value for the reaction of cysteine with OPA is much lower than those of the other amino acids. When cysteine, lysine and proline are excluded from the calculation, an average ϵ value of $6120 \pm 410 \text{ mol}^{-1} \text{ cm}^{-1}$ can be calculated for the reaction of the OPA reagent with amino acids. This value agrees well with ϵ values of 5992 ± 144 , 6420 ± 200 , 6830 ± 350 and $6590 \pm 194 \text{ mol}^{-1} \text{ cm}^{-1}$ as calculated by Church et al. (1983), Frister, Meisel, and Schlimme (1988), Garcia Alvarez-Coque et al. (1989a, b) and Turgeon et al. (1991), respectively. The ϵ value for the reaction of OPA with cysteine ($1330 \pm 130 \text{ mol}^{-1} \text{ cm}^{-1}$) is approximately 20% of this average value (Table 1). The values presented here agree well with ϵ values of 1550 ± 70 and

Table 1

Extinction coefficients for the reaction of the 20 most common amino acids with TNBS and with OPA

Amino acid	OPA	TNBS
	$\epsilon \text{ (mol}^{-1} \text{ cm}^{-1})$	
Alanine	6160 ± 140	12350 ± 260
Arginine	6570 ± 260	10990 ± 160
Asparagine	5870 ± 200	12270 ± 480
Aspartic acid	6010 ± 180	10350 ± 80
Cysteine	1330 ± 130	11800 ± 540
Glutamic acid	6020 ± 240	11900 ± 240
Glutamine	6490 ± 150	13540 ± 1270
Glycine	5870 ± 180	13400 ± 490
Histidine	5890 ± 170	14660 ± 1140
Isoleucine	6350 ± 120	14630 ± 500
Leucine	6750 ± 490	11790 ± 70
Lysine	11420 ± 190	18780 ± 1160
Methionine	6250 ± 780	12180 ± 1400
Phenylalanine	6100 ± 130	11430 ± 950
Proline	—	—
Serine	6550 ± 80	14190 ± 530
Threonine	6320 ± 80	11450 ± 770
Tryptophan	5730 ± 340	9900 ± 280
Tyrosine	4980 ± 590	10160 ± 350
Valine	6130 ± 120	13780 ± 1940

$1380 \pm 346 \text{ mol}^{-1} \text{ cm}^{-1}$ for the reaction of cysteine with OPA reported by Svedas et al. (1980) and Turgeon et al. (1991), respectively. A value of $2700 \text{ mol}^{-1} \text{ cm}^{-1}$ was reported by Garcia Alvarez-Coque et al. (1989a, b) and Medina Hernandez et al. (1990a, b). In comparison, the ϵ value for the reaction of cysteine with TNBS was $11800 \pm 540 \text{ mol}^{-1} \text{ cm}^{-1}$, with the average value for the reaction of all amino acids (excluding cysteine, lysine and proline) calculated as $12290 \pm 1530 \text{ mol}^{-1} \text{ cm}^{-1}$ (Table 1). The reaction of cysteine with TNBS is, therefore comparable to the reaction of the other amino acids with TNBS. Proline and lysine were excluded from calculations as proline does not contain a primary amino group, and therefore does not react with OPA or TNBS, and lysine contains two primary amino groups, therefore reacting more strongly with both reagents.

The instability of the isoindole formed by the reaction of OPA with cysteine may also pose a problem when using the OPA reagent to measure the DH of cysteine-rich substrates. When analysing whey protein hydrolysates, it was found that the absorbance values obtained after addition of the hydrolysate sample to the OPA reagent were unstable, with values increasing over time. This increase in absorbance was more pronounced with hydrolysed, rather than with intact, whey protein. This effect would be expected if the increase was due to cysteine residues, as more of these residues would be exposed in the hydrolysed sample than in the unhydrolysed sample. Fig. 3 shows the stability of absorbance values for cysteine, intact

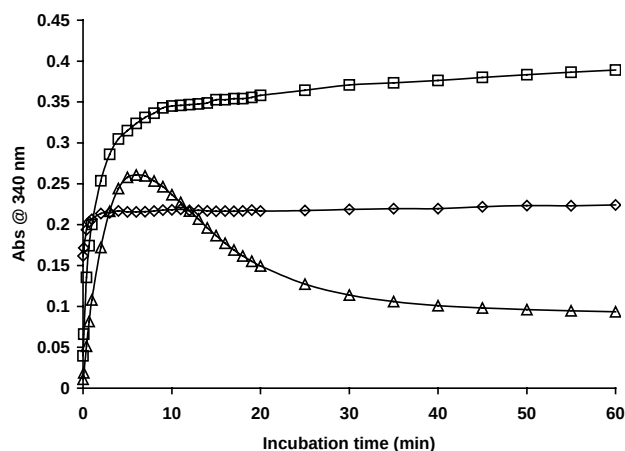


Fig. 3. The stability of absorbance values for cysteine (Δ), unhydrolysed WPC ($\diamond$) and Debitrase HYW20 hydrolysed WPC ($\square$) with the OPA reagent over a 1 h period.

WPC and hydrolysed WPC on incubation with the OPA reagent over a 1 h period.

There was a sharp increase in absorbance during the first 5 min after addition of cysteine to the OPA reagent, after which the absorbance reading levelled off briefly before decreasing rapidly (Fig. 3). This instability of the isoindole formed between cysteine and OPA may be responsible for the instability of the reaction between WPC hydrolysates and OPA. However, the trend shown by hydrolysed WPC was an increase in absorbance over the 1 h period.

4. Conclusion

The pH stat method was found to be a quick and simple method for quantifying the DH of whey protein hydrolysates. The method is non-denaturing and allows for real-time monitoring of the hydrolysis reaction as it proceeds. However, the accuracy of DH values obtained by the pH stat method depends on the type of enzyme activity used in the hydrolysis reaction. When the enzyme preparation is rich in exopeptidase activities, the pH stat method under-estimates DH. The OPA method is also a quick and simple method for the quantification of DH, without long incubation steps, which again allows for real-time monitoring of DH. However, the accuracy of DH values obtained by the OPA method may depend on the protein substrate being hydrolysed. A weak and unstable reaction between OPA and cysteine appears to render the OPA method unsuitable for quantifying DH in cysteine rich substrates such as WPC. The TNBS method proved to be an excellent method for quantifying DH, regardless of the type of enzyme activity used for hydrolysis. However, the TNBS method requires long incubation and cooling steps (1 h and 30 min, respectively) with the result

that the assay cannot be used for real-time monitoring of DH.

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